

# QueryBook — Drawings, Reference Numerals & Brief Description of the Drawings

**Scope:** the complete, current QueryBook figure set — FIG. 1–FIG. 34 — as one unified set: the integral language & speech subsystem (LEL; FIG. 1–8, 31–32), the QueryBook platform (FIG. 9–30), and the Language & Identity Linking (LIL) voiceprint & identity subsystem (FIG. 33–34). **Companion** to the consolidated *QueryBook Provisional Patent Application*; the figures, reference numerals, and brief descriptions here are identical to those in that application. **Document date:** October 1, 2026. Draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman before filing.

## Brief Description of the Drawings

**FIG. 1** is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

**FIG. 2** is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

**FIG. 3** is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

**FIG. 4** is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

**FIG. 5** is a block diagram of the seven-stage Speech Output Pipeline.

**FIG. 6** is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

**FIG. 7** is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

**FIG. 8** is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

## Brief Description of the QueryBook Platform Drawings (FIG. 9–FIG. 30)

**FIG. 9** is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

**FIG. 10** is a diagram of the fact Unit — atom, envelope, and content-derived identity.

**FIG. 11** is a diagram of the fact-certification authority hierarchy.

**FIG. 12** is a diagram of the uFCS substrate — three-store architecture.

**FIG. 13** is a diagram of the authored tour and exhibition sequencing.

**FIG. 14** is a diagram of the fQL — joint structure and trust execution.

**FIG. 15** is a diagram of the verifying machine-generated text against certified records.

**FIG. 16** is a diagram of the federation — cross-boundary re-validation.

**FIG. 17** is a diagram of the fact-level erasure by external-key destruction.

**FIG. 18** is a diagram of the universal security architecture.

**FIG. 19** is a diagram of the universal honeypot saturation and data firehose.

**FIG. 20** is a diagram of the deployment topologies over an invariant common layer.

**FIG. 21** is a diagram of the adaptive response engine — directive fixed before generation.

**FIG. 22** is a diagram of the user-owned agent system with governance and kill switch.

**FIG. 23** is a diagram of the deterministic multi-language regeneration.

**FIG. 24** is a diagram of the semantic compression, expansion, colorization and upscaling.

**FIG. 25** is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

**FIG. 26** is a diagram of the neuromorphic cognitive layer — attractor convergence.

**FIG. 27** is a diagram of the fact Unit — confidence and evidence model.

**FIG. 28** is a diagram of the uFCS packet structure — serialized Fact Unit record.

**FIG. 29** is a diagram of the uFCS packet framing — content-addressed block storage.

**FIG. 30** is a diagram of the fQL — query anatomy and worked example.

## **Reference Numerals**

| No. | Element                                        | No.  | Element                                                   |
|-----|------------------------------------------------|------|-----------------------------------------------------------|
| 100 | QueryBook LEL system                           | 764  | Phase-4 agent (declines — spec-only)                      |
| 110 | Sub-Language Priming Layer (SLPL)              | 800  | deterministic Phase-Two semantic grounding                |
| 120 | Language Expression Layer (LEL)                | 810  | Phase-One vocabulary (learned words)                      |
| 130 | Transition Gate                                | 820  | deterministic grounder (per word)                         |
| 140 | Phase Two semantic-communicative learning      | 830  | bundled public-domain dictionary source                   |
| 150 | Multilingual Delta Acquisition module          | 831  | dictionary meaning Fact Unit (means, trust 0.9)           |
| 160 | Speech Output Pipeline                         | 840  | self-grounding store-lookup source                        |
| 165 | natural speech output                          | 841  | store-link Fact Unit (grounded_by_fact)                   |
| 170 | co-grounding interface                         | 850  | Fact Unit store (provenance, domain language)             |
| 180 | Federated Query Language (FQL)                 | 860  | excluded large-language-model meaning source              |
| 190 | Universal Federated Connectivity System (UFCS) | 870  | optional future dictionary-gated suggestor                |
| 195 | federated data sources                         | 900  | advanced multilingual pronunciation subsystem             |
| 200 | SLPL                                           | 910  | pronunciation request (word/phrase, language)             |
| 210 | acoustic phonetic processing                   | 920  | pronunciation preference-ladder selector                  |
| 211 | signal-level feature embedding                 | 930  | bundled open-source phonemizer                            |
| 220 | phonics / phonemic awareness                   | 931  | phonemizer IPA transcription                              |
| 221 | phoneme-boundary segmenter                     | 932  | phonemizer call cache                                     |
| 230 | communication-science foundations              | 933  | bounded call log / timeout guard                          |
| 231 | communicative-salience weight                  | 940  | rule-seeded G2P tables (per language)                     |
| 240 | prosody & rhythm processing                    | 941  | approximate rule-seeded IPA (offline)                     |
| 241 | prosodic contour vector                        | 950  | syllable count + primary-stress derivation                |
| 250 | information-theoretic grounding                | 960  | explicit null result (not fabricated)                     |
| 251 | information weighting                          | 970  | phrase-level single-call transcription                    |
| 260 | neurolinguistic chunking                       | 971  | displayed transcription consistent with synthesized audio |
| 261 | chunk segments                                 | 980  | self-contained application bundle                         |
| 270 | licensed multimedia input                      | 981  | bundled open-source speech engine                         |
| 300 | developmental learning sequence                | 982  | runtime-support libraries                                 |
| 310 | Phase One (structural-acoustic)                | 983  | bundled phoneme data                                      |
| 311 | sound discrimination                           | 984  | engine-selection ladder (per OS / language)               |
| 312 | syllabic segmentation                          | 985  | host OS built-in voice service                            |
| 313 | word-as-acoustic-object                        | 986  | culture-matched OS voice (when installed)                 |
| 314 | semantics suppressed                           | 987  | honest refusal with remedy (no fabricated audio)          |
| 320 | Transition Gate (one-way)                      | 990  | deterministic translator (through English)                |
| 321 | phonemic-inventory completeness $\geq \tau_1$  | 991  | bundled open-source bilingual dictionary                  |
| 322 | lexical stability $\geq \tau_2$                | 992  | coverage accounting + unknown-word passthrough (flagged)  |
| 323 | co-occurrence density $\geq \tau_3$            | 993  | deterministic language detector                           |
| 324 | pattern saturation $< \epsilon$                | 995  | permanent quality-control battery                         |
| 330 | Phase Two (semantic-communicative)             | 996  | IPA-produced check                                        |
| 331 | word-to-referent grounding                     | 997  | syllable-sanity check ( $\pm 1$ )                         |
| 332 | communicative discovery                        | 998  | valid-waveform synthesis check                            |
| 333 | emergent grammar / syntax                      | 999  | per-language QC report + overall pass score               |
| 400 | Delta Acquisition Model                        | 1000 | LIL voiceprint & identity subsystem                       |
| 410 | reused L1 foundation                           | 1001 | consent gate (precondition)                               |
| 420 | L2 delta detector                              | 1002 | audio/video capture + VAD/segmentation                    |
| 430 | targeted training of novel elements            | 1003 | acoustic feature extraction                               |
| 440 | multilingual semantic alignment                | 1004 | speaker embedding (voiceprint vector)                     |
| 450 | acquisition acceleration flywheel              | 1005 | enrollment — canonical voiceprint                         |
| 451 | negative-transfer detection & retraining       | 1006 | match / score (cosine · PLDA)                             |
| 500 | Speech Output Pipeline                         | 1007 | QueryBook trust gate<br>(VERIFIED/CONTRADICTED/UNKNOWN)   |
| 510 | lexical selection                              | 1008 | Fact-Unit identity graph                                  |
| 520 | grapheme-to-phoneme conversion                 | 1009 | LEL cross-script name/entity resolution                   |
| 530 | coarticulation modeling                        | 1010 | provenance-tracked link Fact Unit                         |
| 540 | prosody generation                             | 1020 | voice authentication request                              |
| 550 | register & affect calibration                  | 1021 | capture phrase + embed                                    |
| 560 | self-monitoring feedback loop                  | 1022 | anti-spoofing (liveness/replay/synthetic)                 |
| 570 | neural vocoder rendering                       | 1023 | deterministic risk-scoring policy                         |

| No. | Element |
|-----|---------|
|-----|---------|

|     |                                                                          |
|-----|--------------------------------------------------------------------------|
| 580 | natural speech waveform                                                  |
| 600 | co-grounded integration                                                  |
| 610 | natural-language input                                                   |
| 620 | LEL analysis                                                             |
| 630 | FQL operations                                                           |
| 640 | UFCS execution                                                           |
| 650 | federated sources                                                        |
| 660 | results                                                                  |
| 670 | render to NL / speech in input language                                  |
| 680 | language-agnostic resolution                                             |
| 700 | Language Lab prototype embodiment                                        |
| 710 | corpus source                                                            |
| 720 | Phase-One structural agent                                               |
| 730 | structural analyzer                                                      |
| 731 | grapheme inventory                                                       |
| 732 | rule-seeded G2P                                                          |
| 733 | syllabifier                                                              |
| 734 | lexical-stability metric                                                 |
| 735 | co-occurrence density metric                                             |
| 736 | prosody proxy                                                            |
| 740 | Fact Unit store                                                          |
| 750 | Transition-Gate metrics                                                  |
| 751 | one-way latch                                                            |
| 760 | phase agents on shared manager                                           |
| 761 | Phase-1 agent (implemented)                                              |
| 762 | Phase-2 agent — deterministic dictionary + store grounding (implemented) |
| 763 | Phase-3 agent (declines — spec-only)                                     |

| No. | Element |
|-----|---------|
|-----|---------|

|      |                                                                            |
|------|----------------------------------------------------------------------------|
| 1024 | authentication decision gate                                               |
| 1025 | second-factor challenge                                                    |
| 1026 | auth-event Fact Unit (provenance)                                          |
| 1100 | UIAS integrity audit service                                               |
| 1101 | monitored system workflow (unmodified)                                     |
| 1102 | input-boundary position                                                    |
| 1103 | processing-layer position                                                  |
| 1104 | storage-layer position                                                     |
| 1105 | output-boundary / handoff position                                         |
| 1106 | Connector + Ingestion agents (seal at observation)                         |
| 1107 | seal = Fact Unit (hash · id-hash · independent timestamp · source · agent) |
| 1108 | Similarity Detection Agent (semantic signatures)                           |
| 1109 | Fact-Unit audit ledger (UFCS, append-only, immutable)                      |
| 1110 | contemporaneous cross-party anomaly flag                                   |
| 1111 | Verification Agent (continuous re-verification)                            |
| 1112 | content-not-stored / one-way identity (fact-level erasure)                 |
| 1120 | UIAS signed Proof Package generation                                       |
| 1121 | accusation / inquiry                                                       |
| 1122 | Proof Package Agent                                                        |
| 1123 | sealed Fact Units + verified seals                                         |
| 1124 | cross-submission similarity log entry                                      |
| 1125 | chronological ledger extract (RPH)                                         |
| 1126 | Arbitration Agent (factual determination; Prime-Directive gate)            |
| 1127 | assemble + sign under UIAS independent identity                            |
| 1128 | signed Proof Package                                                       |
| 1129 | factual-only determination (legal reserved to courts/regulators)           |

## Drawings

Formal patent drawings — black line art, forming **one QueryBook figure set**: FIG. 1–FIG. 8 and FIG. 31–FIG. 32 depict QueryBook’s integral developmental language and speech subsystem (the Language Expression Layer), FIG. 9–FIG. 30 depict the QueryBook platform in which that subsystem is co-grounded (see FIG. 6), FIG. 33–FIG. 34 depict the Language & Identity Linking (LIL) voiceprint and multimodal identity subsystem, and FIG. 35–FIG. 36 depict the Universal Integrity Audit Service (UIAS) that QueryBook delivers on these primitives. Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

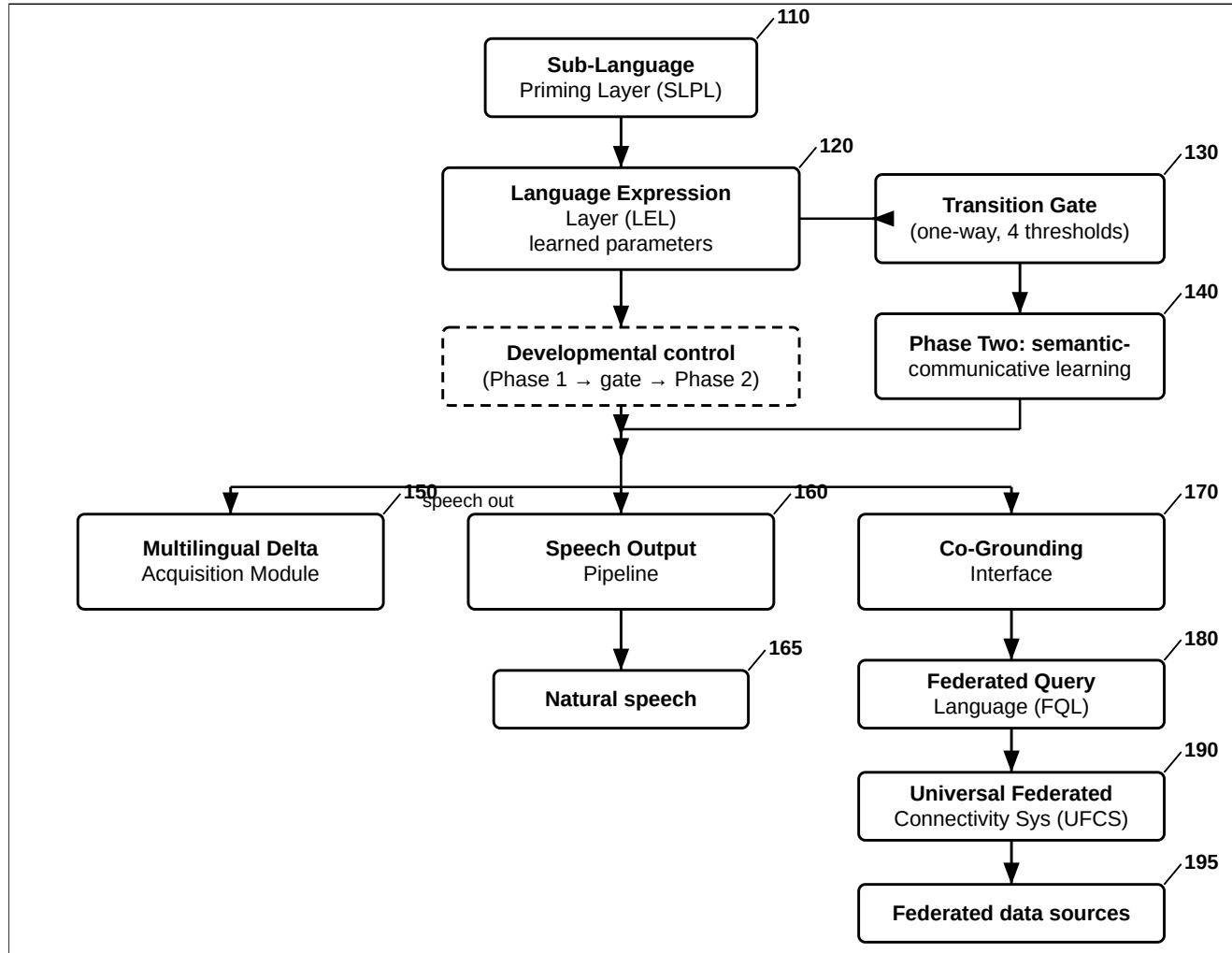


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

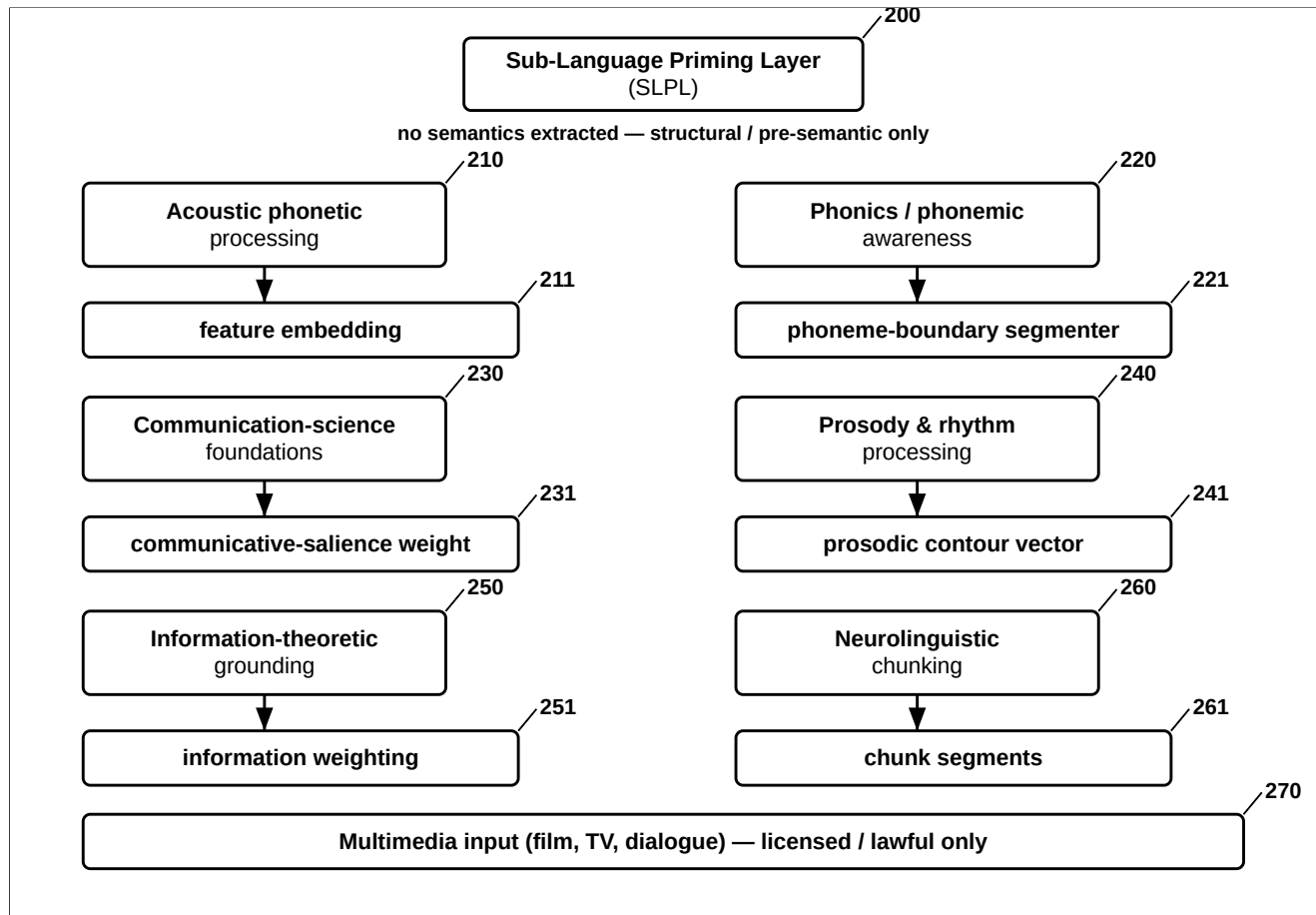


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

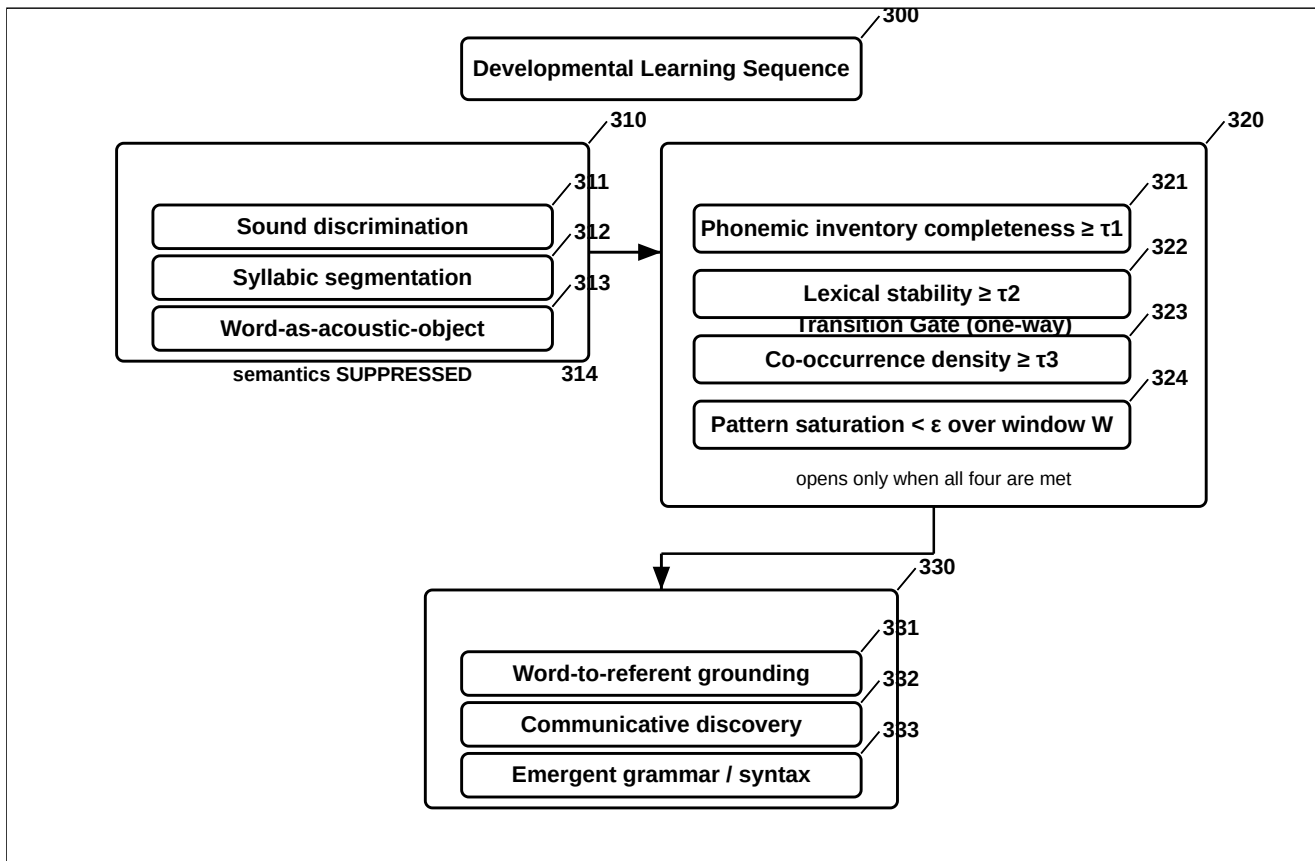


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

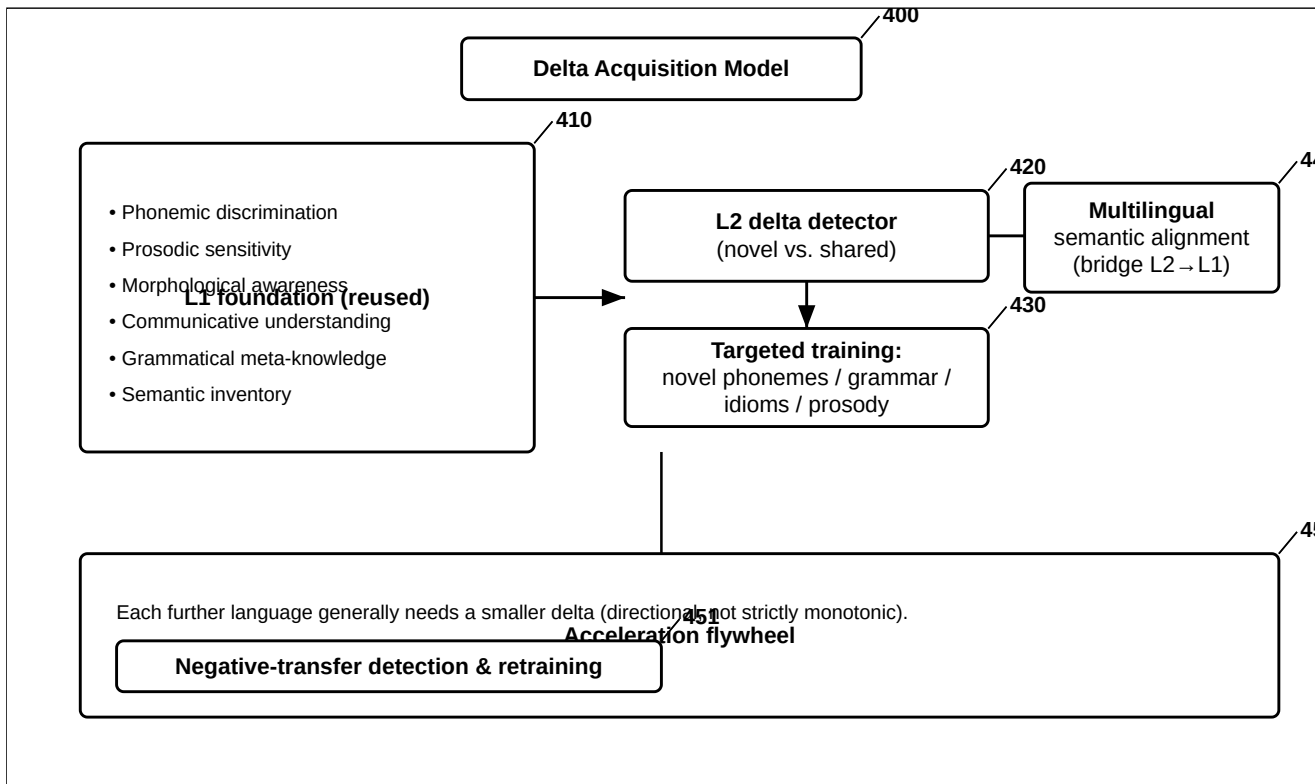


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

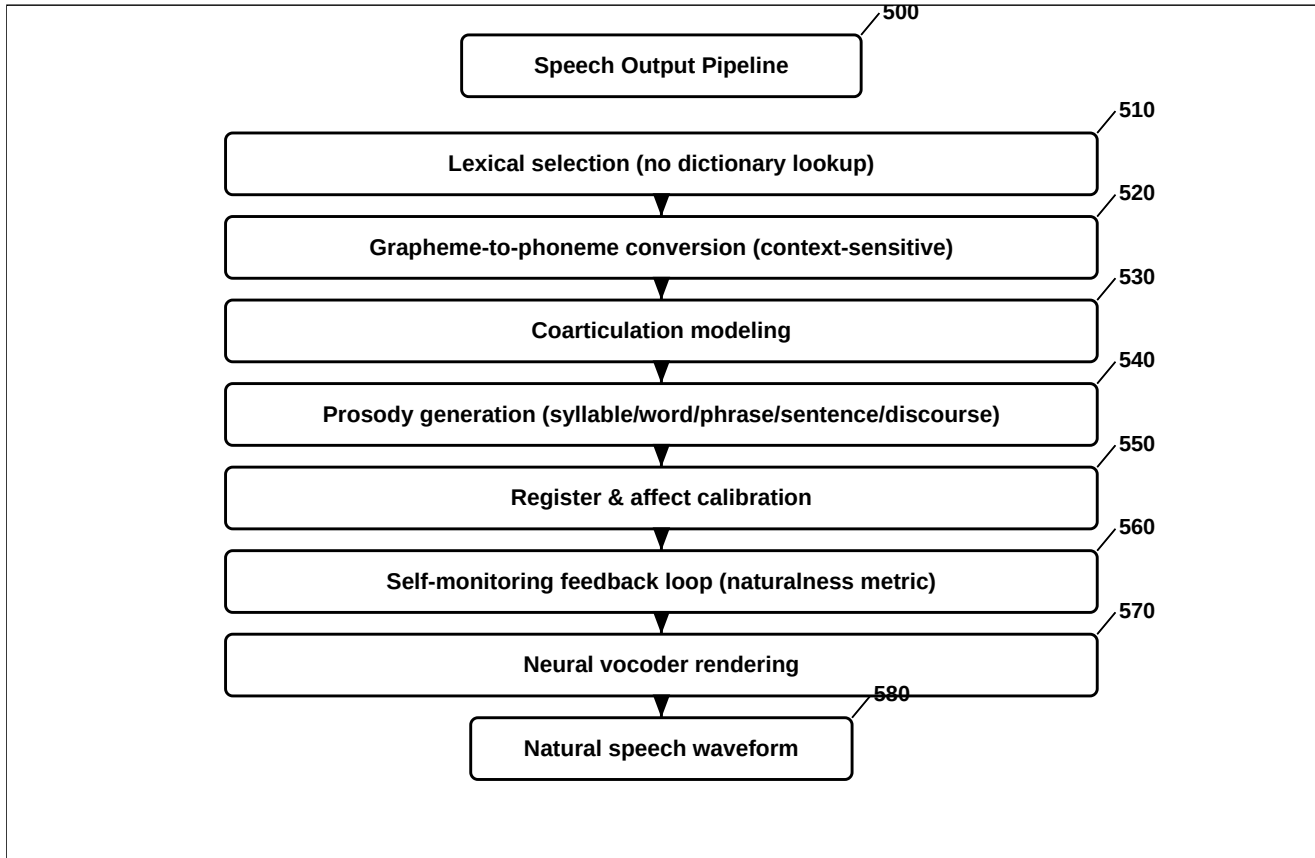


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

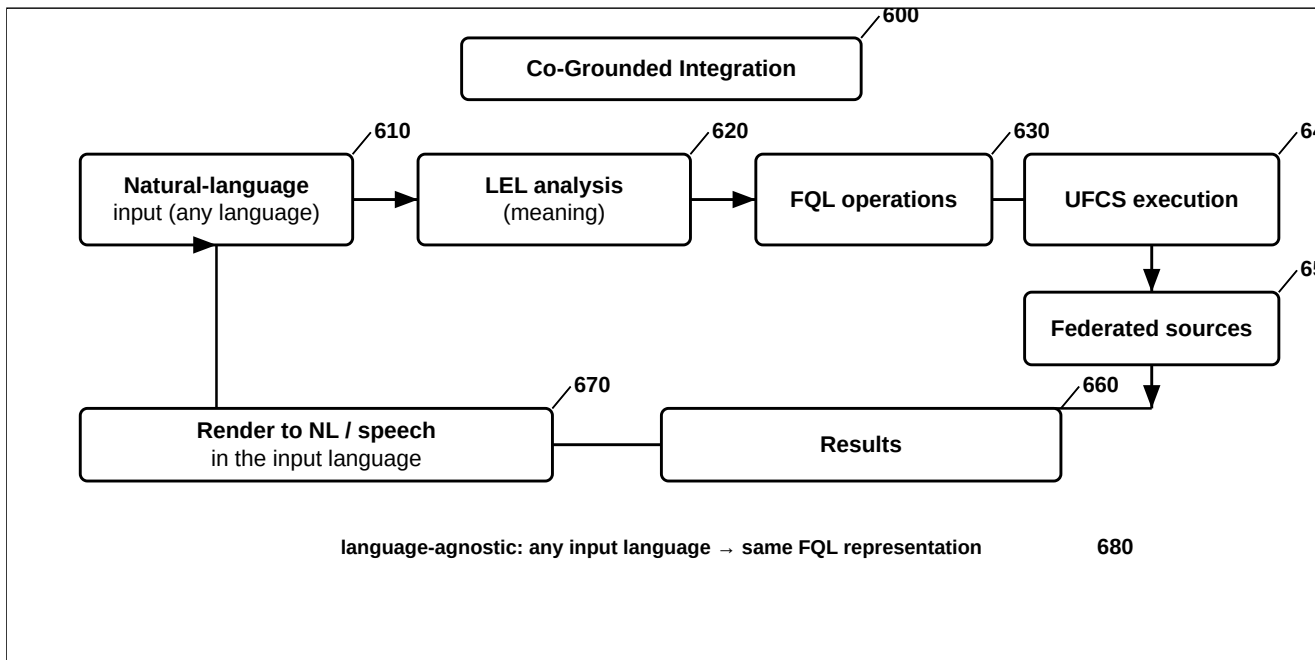


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

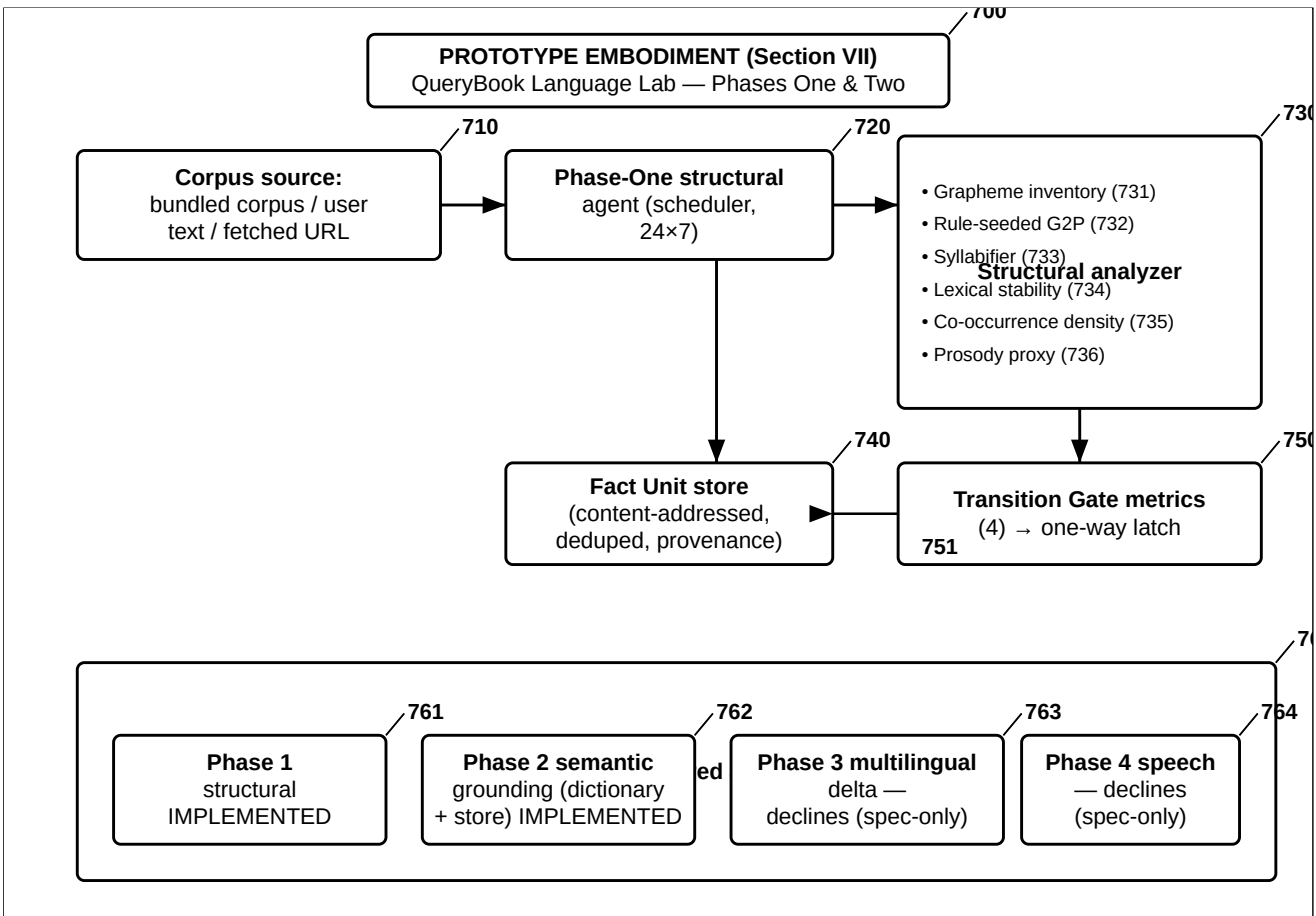


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

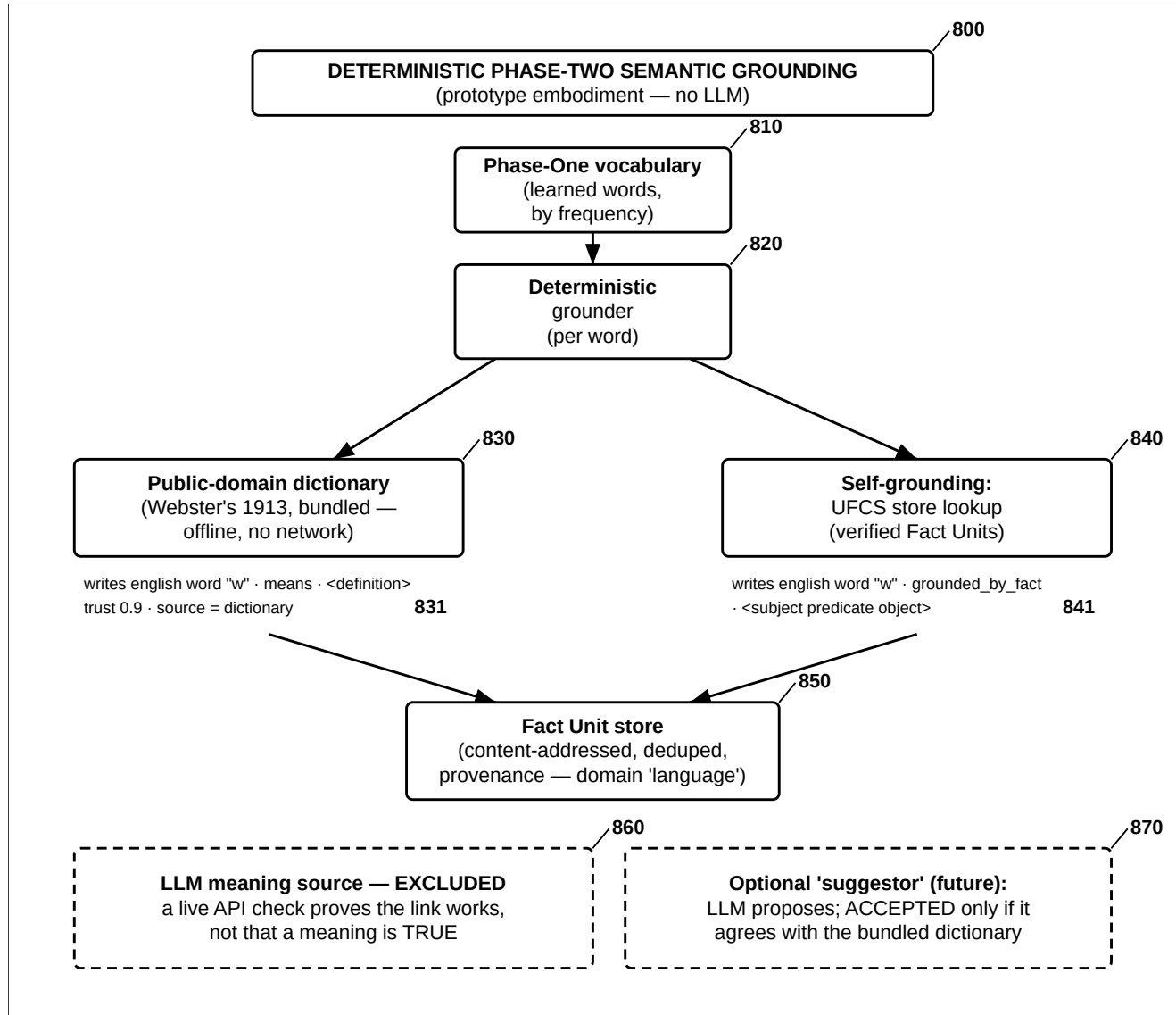


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

## QueryBook Platform Drawings (FIG. 9–FIG. 30) — the system in which the language subsystem is integral

FIG. 9 through FIG. 30 depict the QueryBook deterministic, provenance-tracked federated knowledge and query platform of which the developmental language and speech subsystem (the Language Expression Layer, FIG. 1–FIG. 8 and FIG. 31–FIG. 32) is an integral, co-grounded part (see FIG. 6). Together with the language-subsystem figures these form **one QueryBook figure set**. Draft — the language-subsystem figures use a 100–999 numeral series and the platform figures a 100–2200 series; counsel/draftsperson to consolidate the two into a single QueryBook numeral series and finalize to 37 CFR 1.84 before filing.

FIG. 9

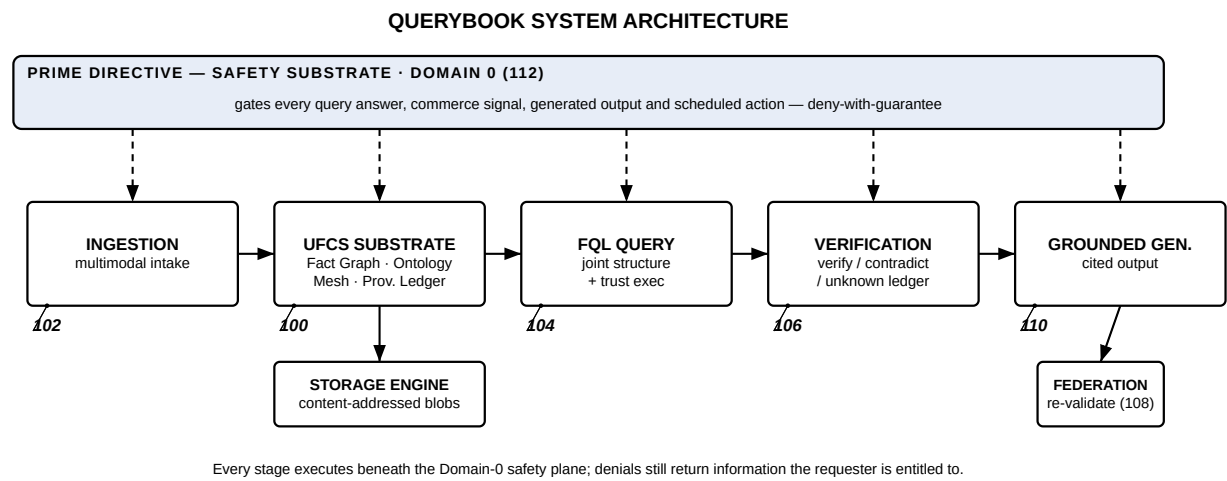


FIG. 9

FIG. 9 — QueryBook system architecture (Prime Directive as Domain 0)

FIG. 10

FACT UNIT — ATOM, ENVELOPE, AND CONTENT-DERIVED IDENTITY

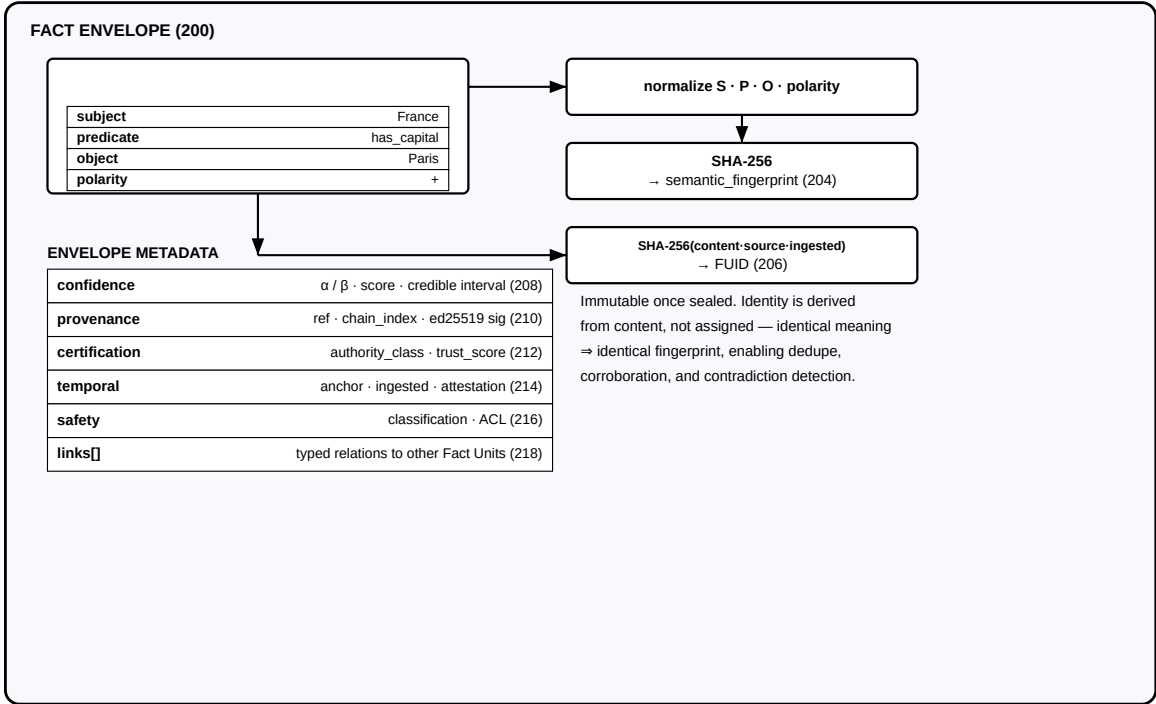


FIG. 10

FIG. 10 — Fact Unit — atom, envelope, and content-derived identity

FIG. 11

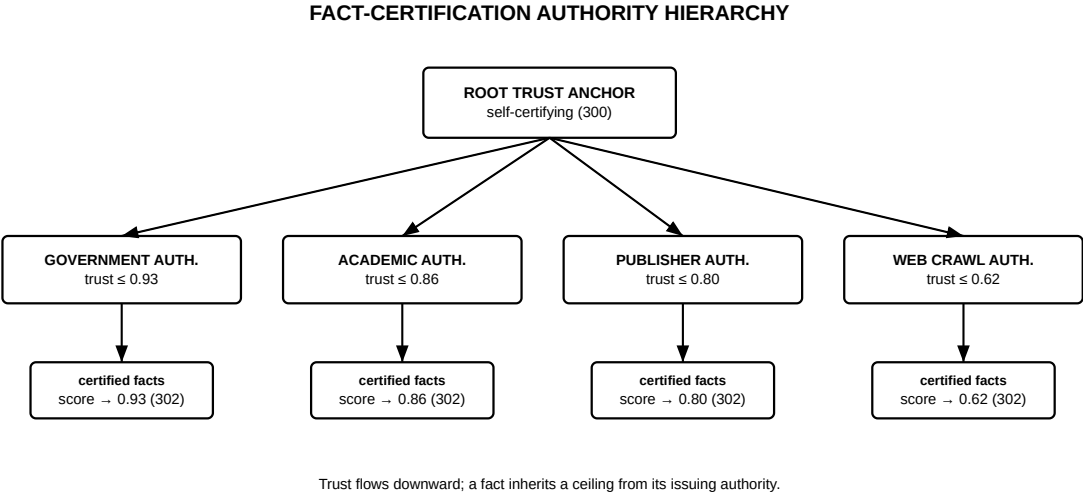


FIG. 11

FIG. 11 — Fact-certification authority hierarchy

FIG. 12

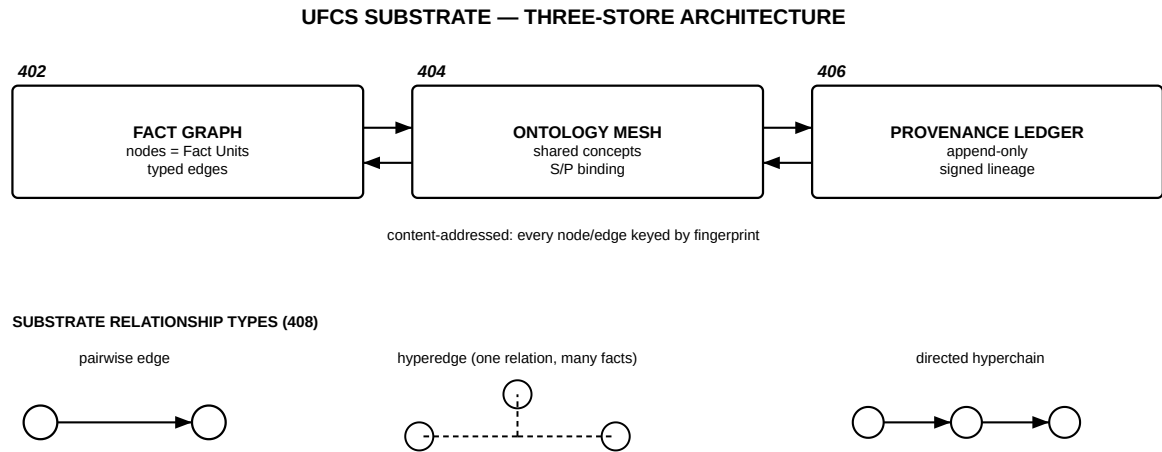
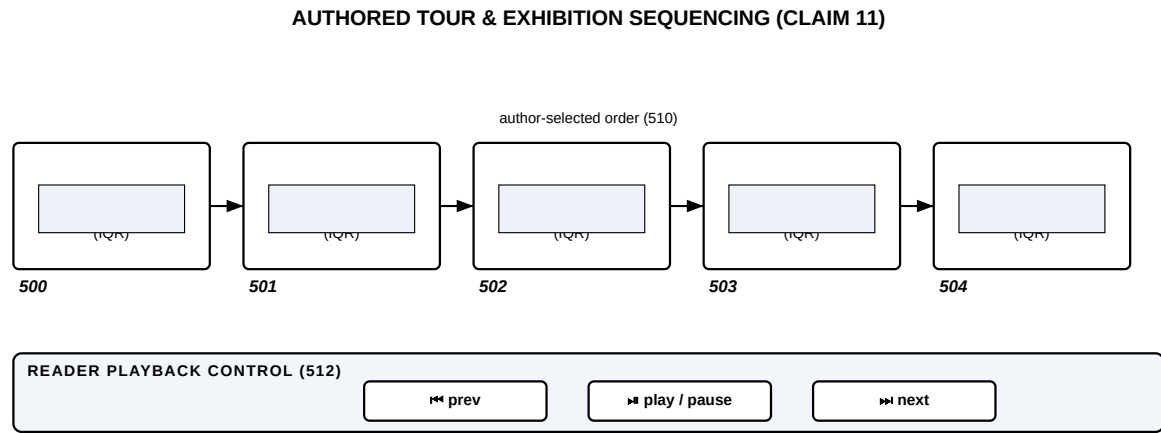


FIG. 12

FIG. 12 — UFCS substrate — three-store architecture

FIG. 13



Author fixes the sequence; the reader controls pace and direction along the ordered waypoints.

FIG. 13

FIG. 13 — Authored tour and exhibition sequencing

FIG. 14

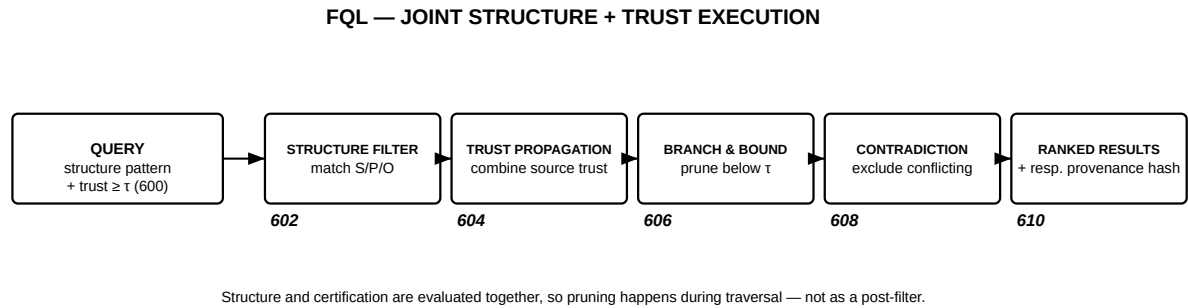


FIG. 14

FIG. 14 — FQL — joint structure and trust execution

FIG. 15

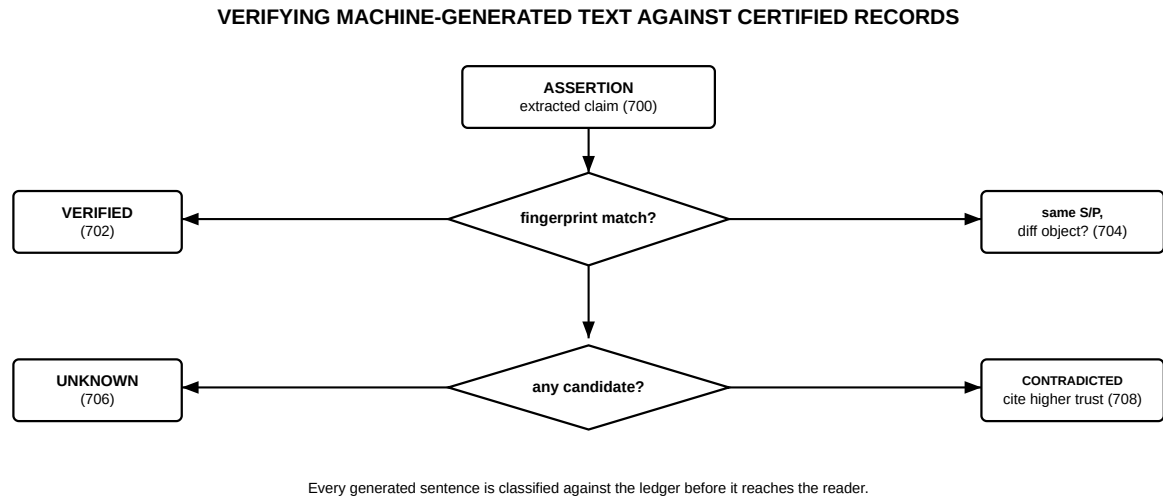


FIG. 15

FIG. 15 — Verifying machine-generated text against certified records

FIG. 16

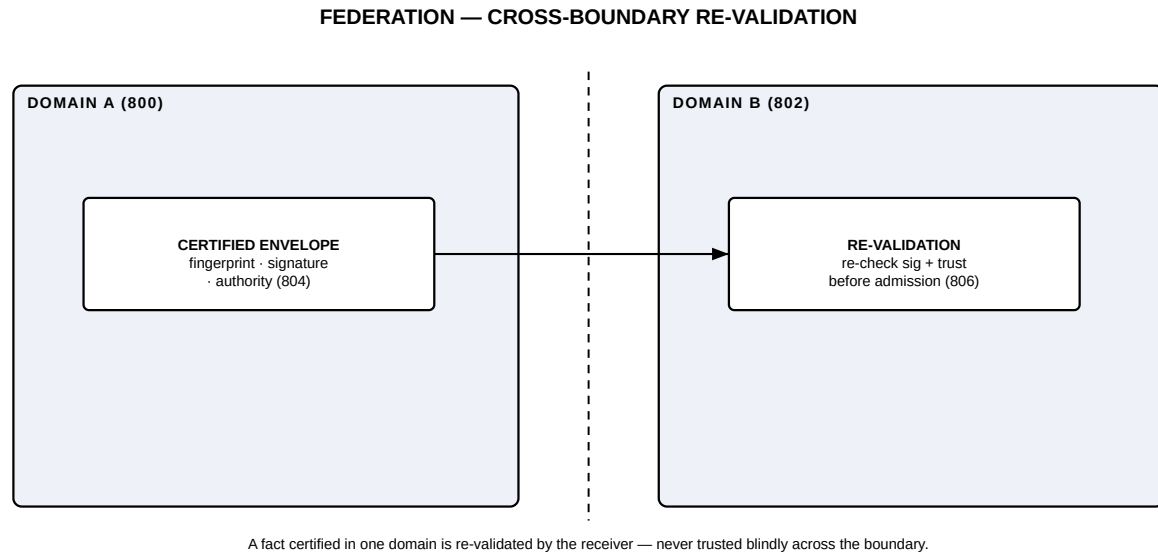
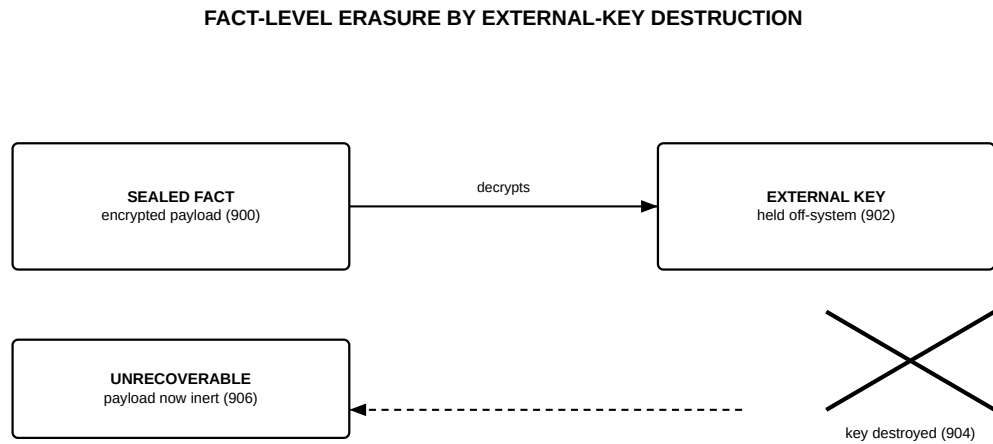


FIG. 16

FIG. 16 — Federation — cross-boundary re-validation

FIG. 17



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 17

FIG. 17 — Fact-level erasure by external-key destruction

FIG. 18

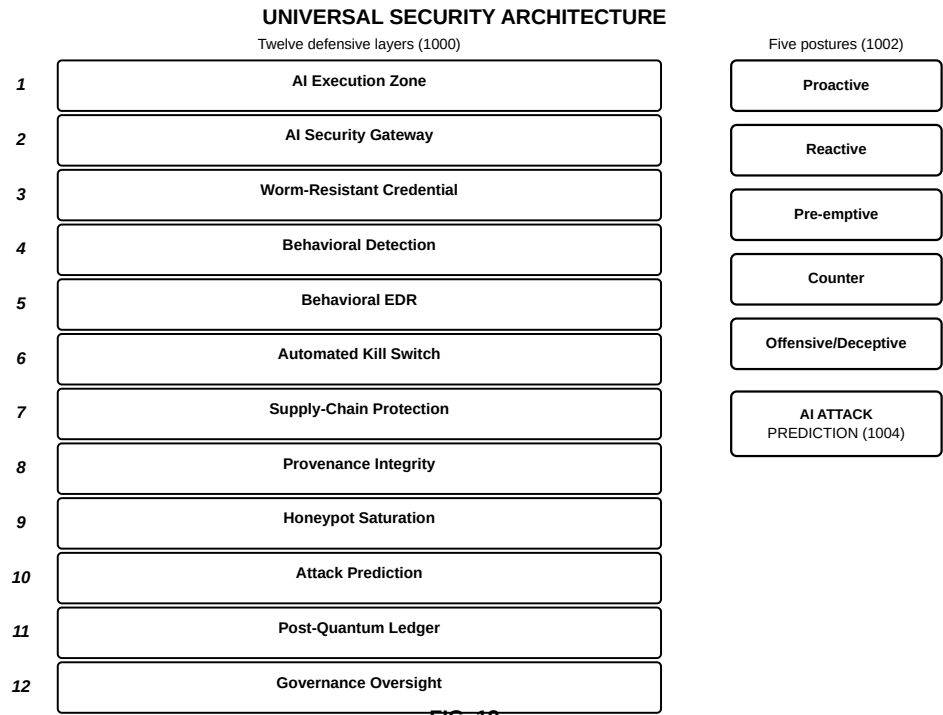


FIG. 18 — Universal security architecture

FIG. 19

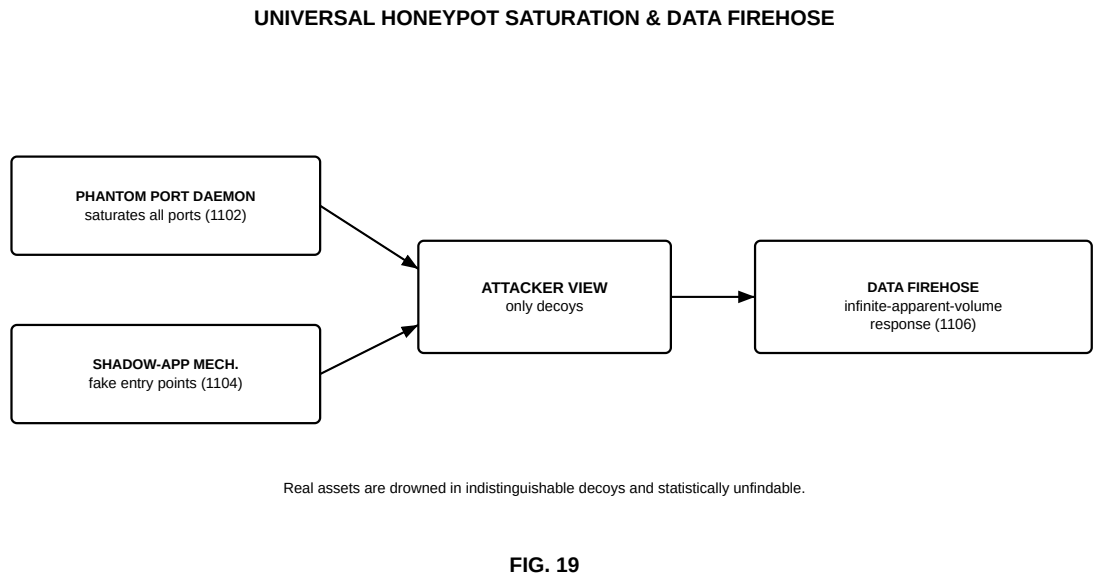


FIG. 19 — Universal honeypot saturation and data firehose

FIG. 20

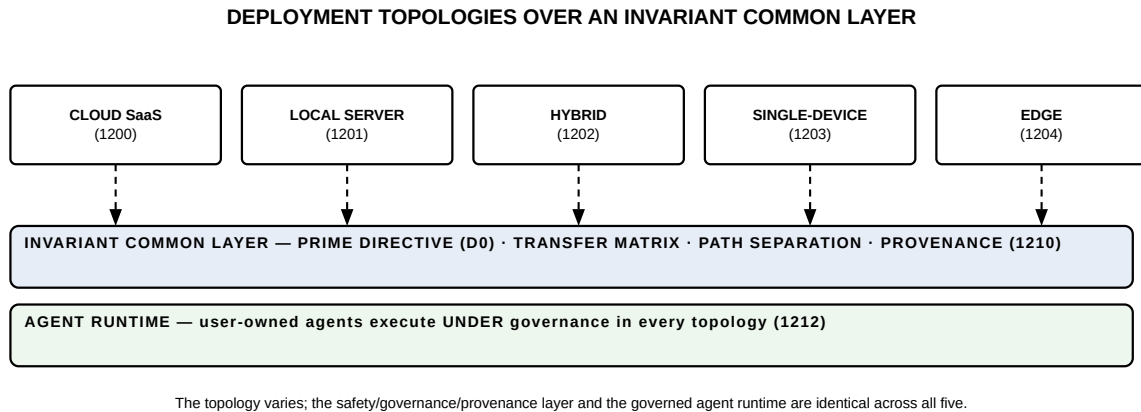


FIG. 20

FIG. 20 — Deployment topologies over an invariant common layer

FIG. 21

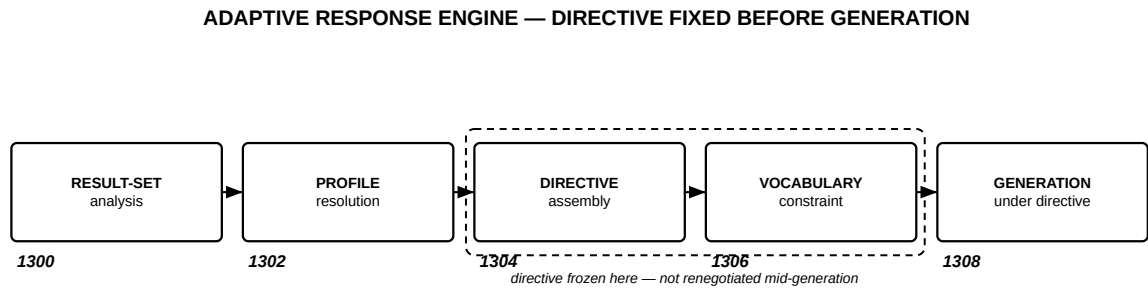
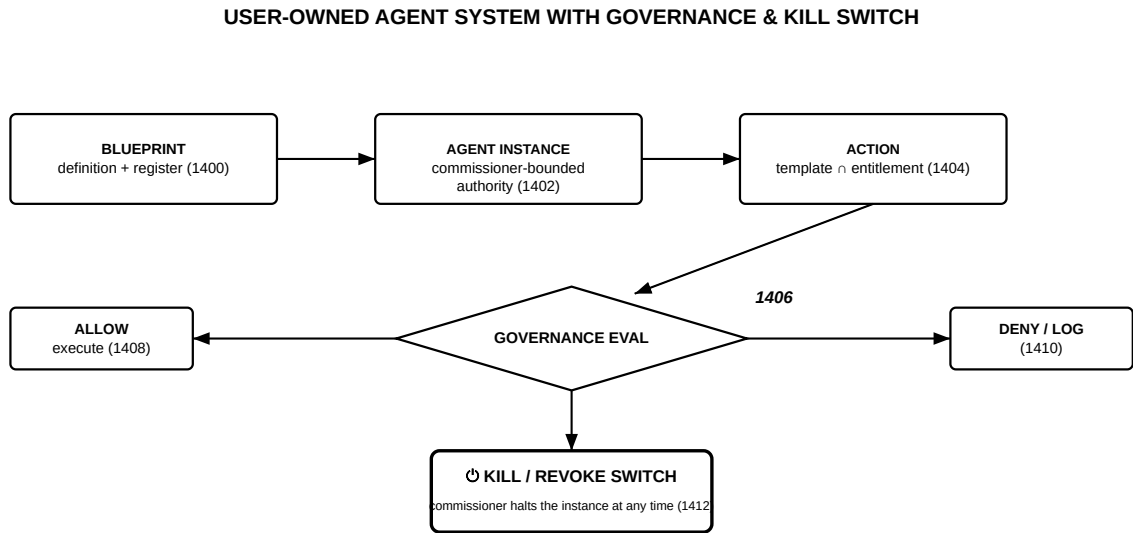


FIG. 21

FIG. 21 — Adaptive response engine — directive fixed before generation

FIG. 22

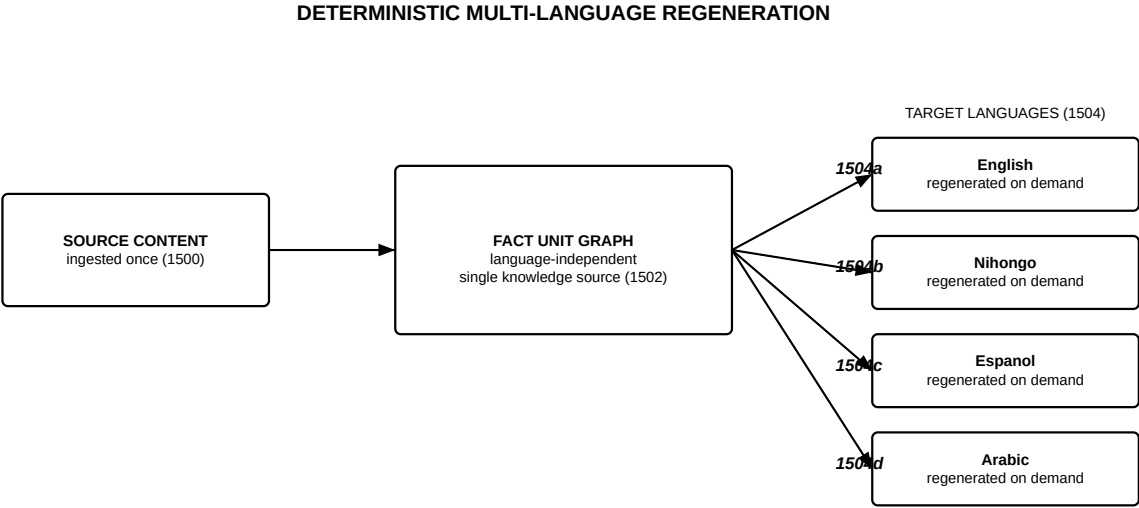


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 22

FIG. 22 — User-owned agent system with governance and kill switch

FIG. 23



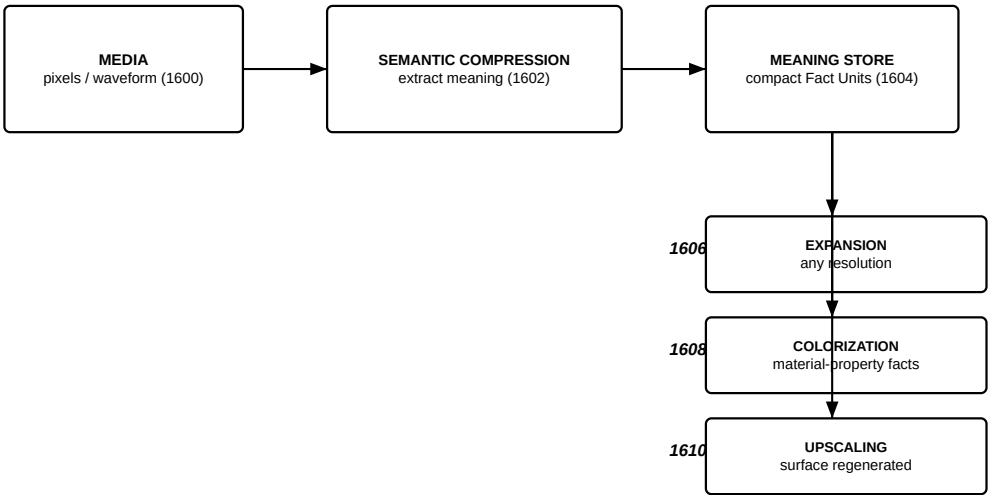
One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 23

FIG. 23 — Deterministic multi-language regeneration

FIG. 24

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING



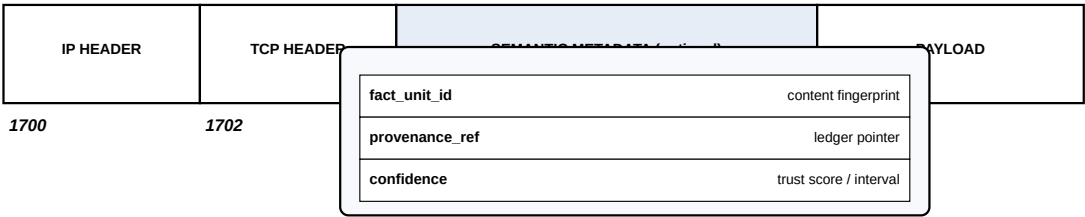
Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 24

FIG. 24 — Semantic compression, expansion, colorization and upscaling

FIG. 25

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 25

FIG. 25 — QueryBook-enhanced TCP/IP hybrid protocol packet

FIG. 26

NEUROMORPHIC COGNITIVE LAYER — ATTRACTOR CONVERGENCE

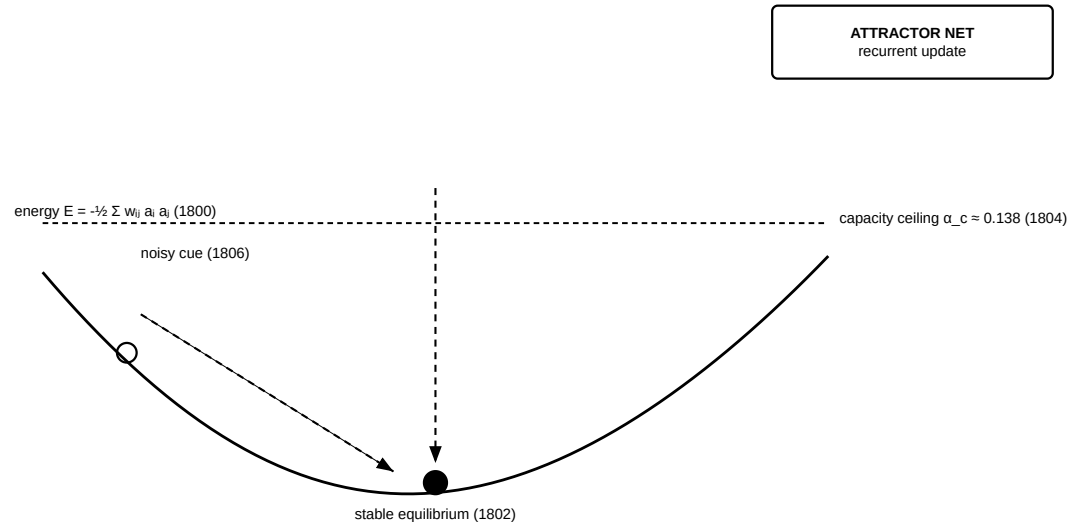
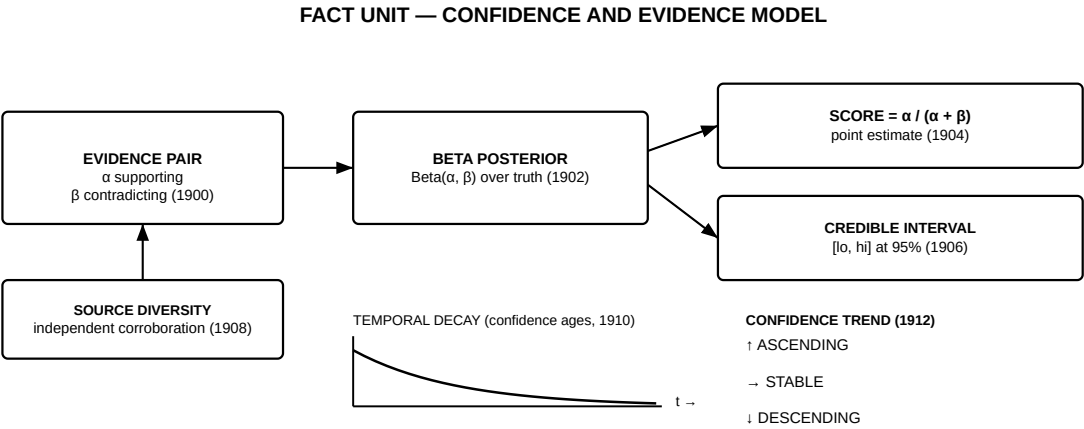


FIG. 26

FIG. 26 — Neuromorphic cognitive layer — attractor convergence

FIG. 27



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 27

FIG. 27 — Fact Unit — confidence and evidence model

FIG. 28

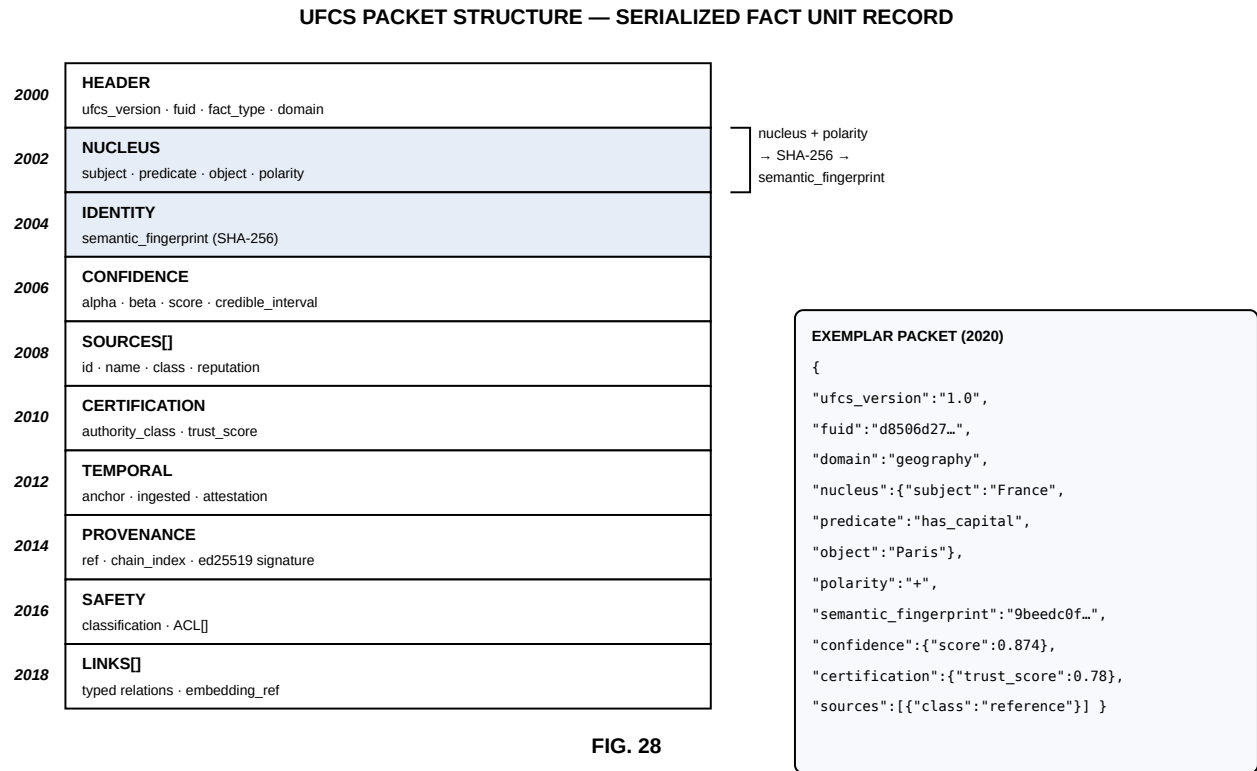


FIG. 28 — UFCS packet structure — serialized Fact Unit record

FIG. 29

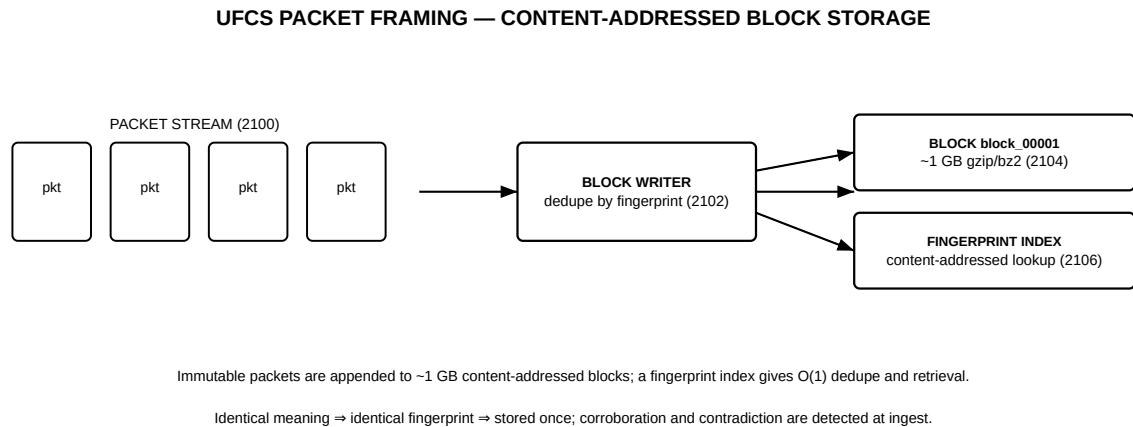


FIG. 29

FIG. 29 — UFCS packet framing — content-addressed block storage

FIG. 30

FQL — QUERY ANATOMY AND WORKED EXAMPLE

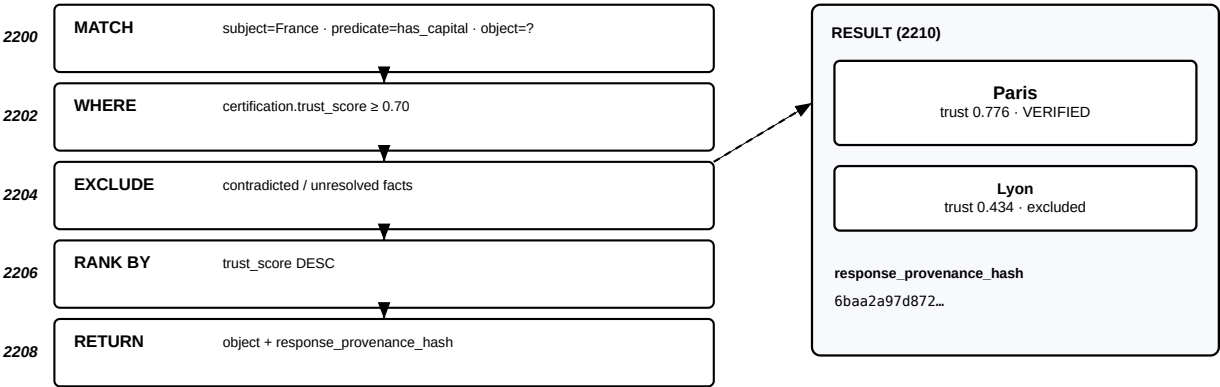


FIG. 30

FIG. 30 — FQL — query anatomy and worked example

FIG. 31

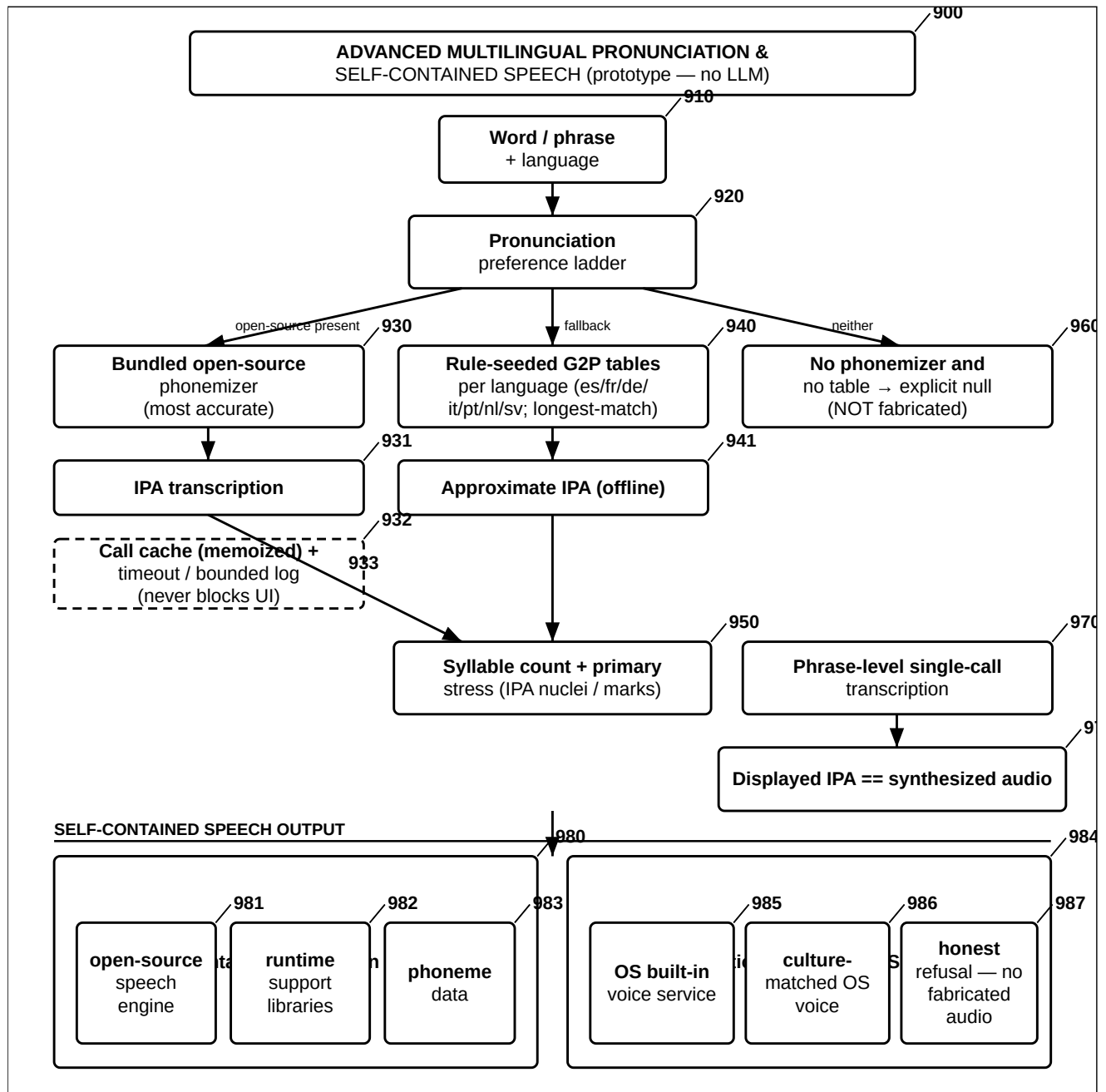


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

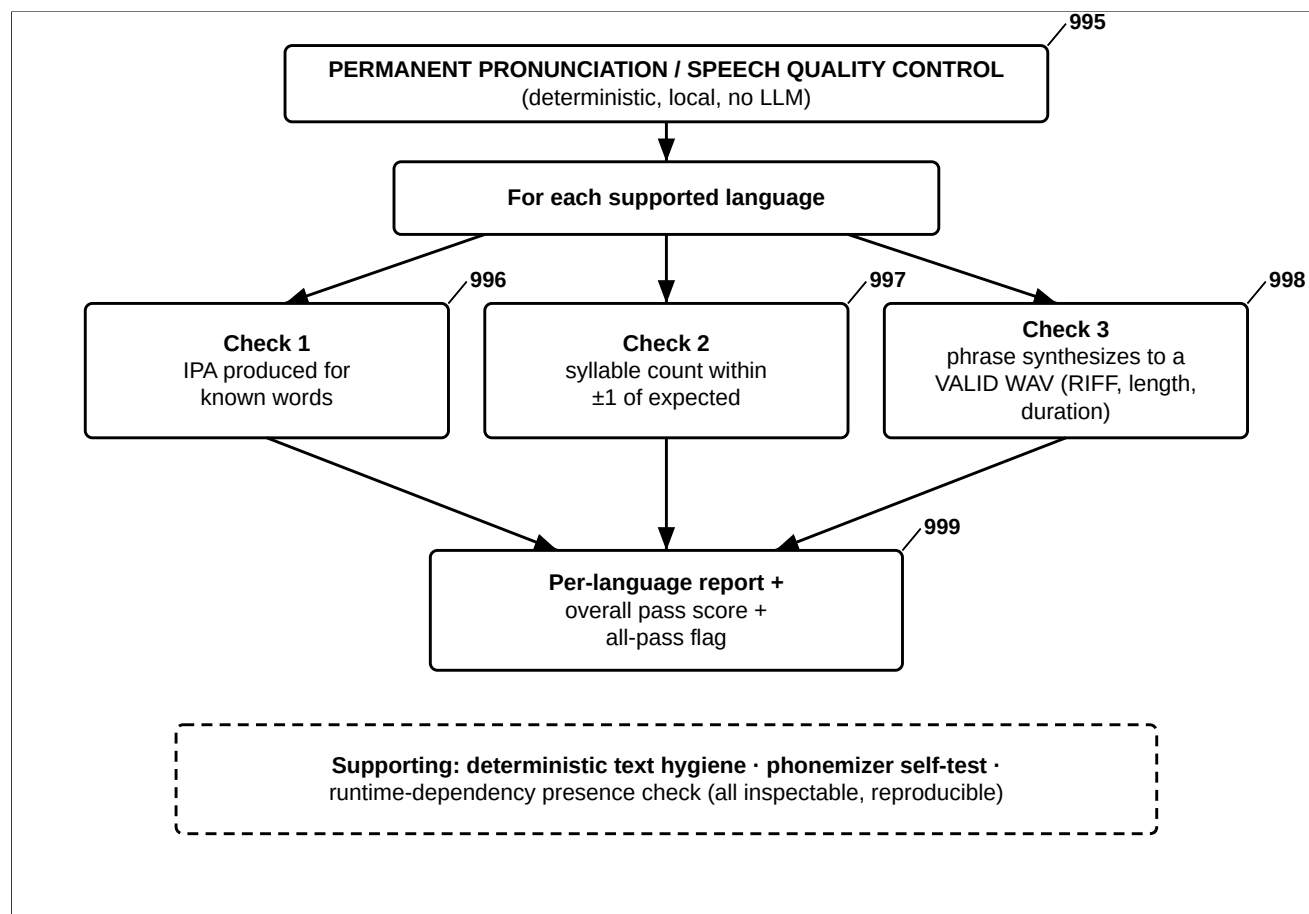


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

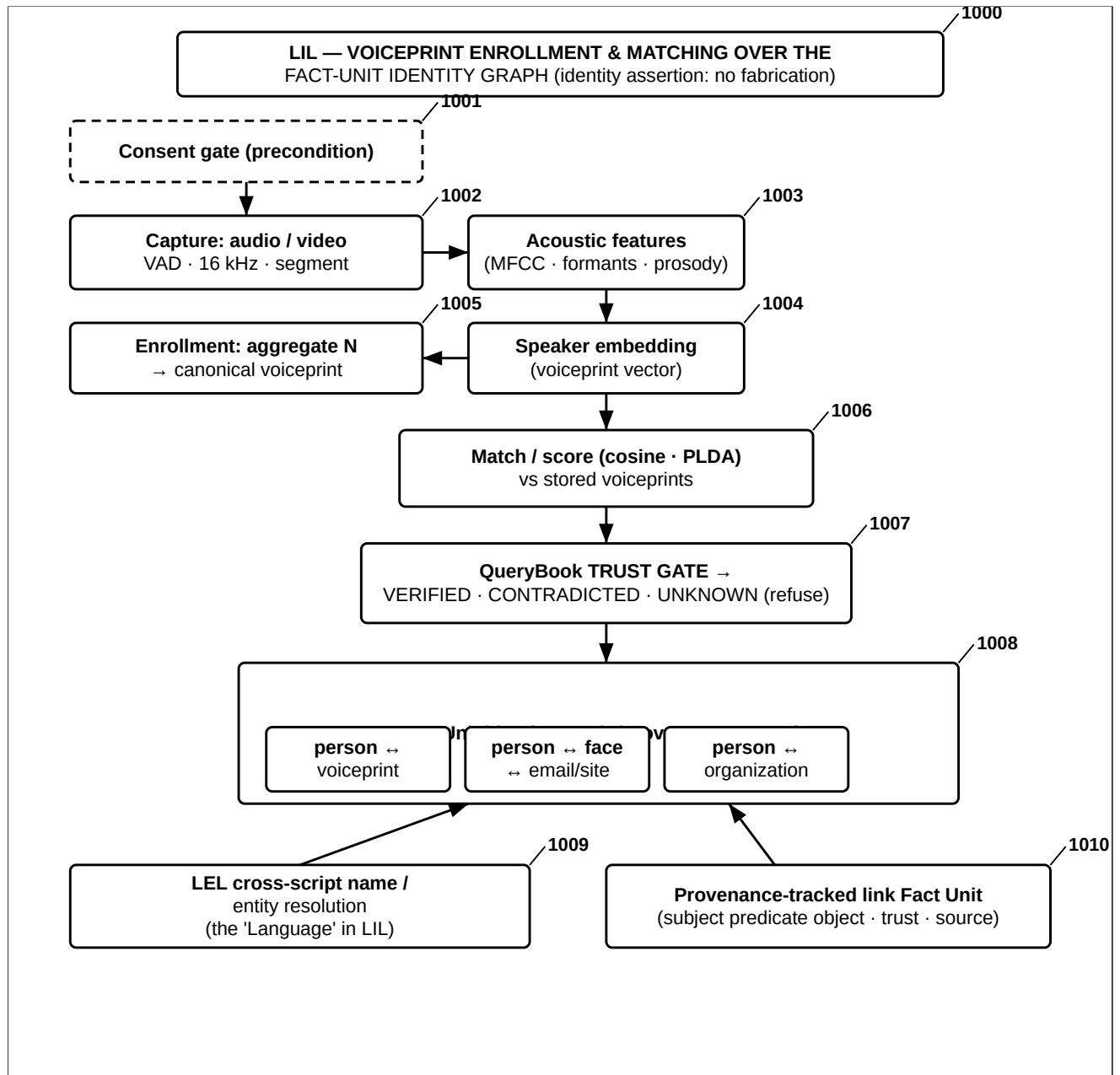


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

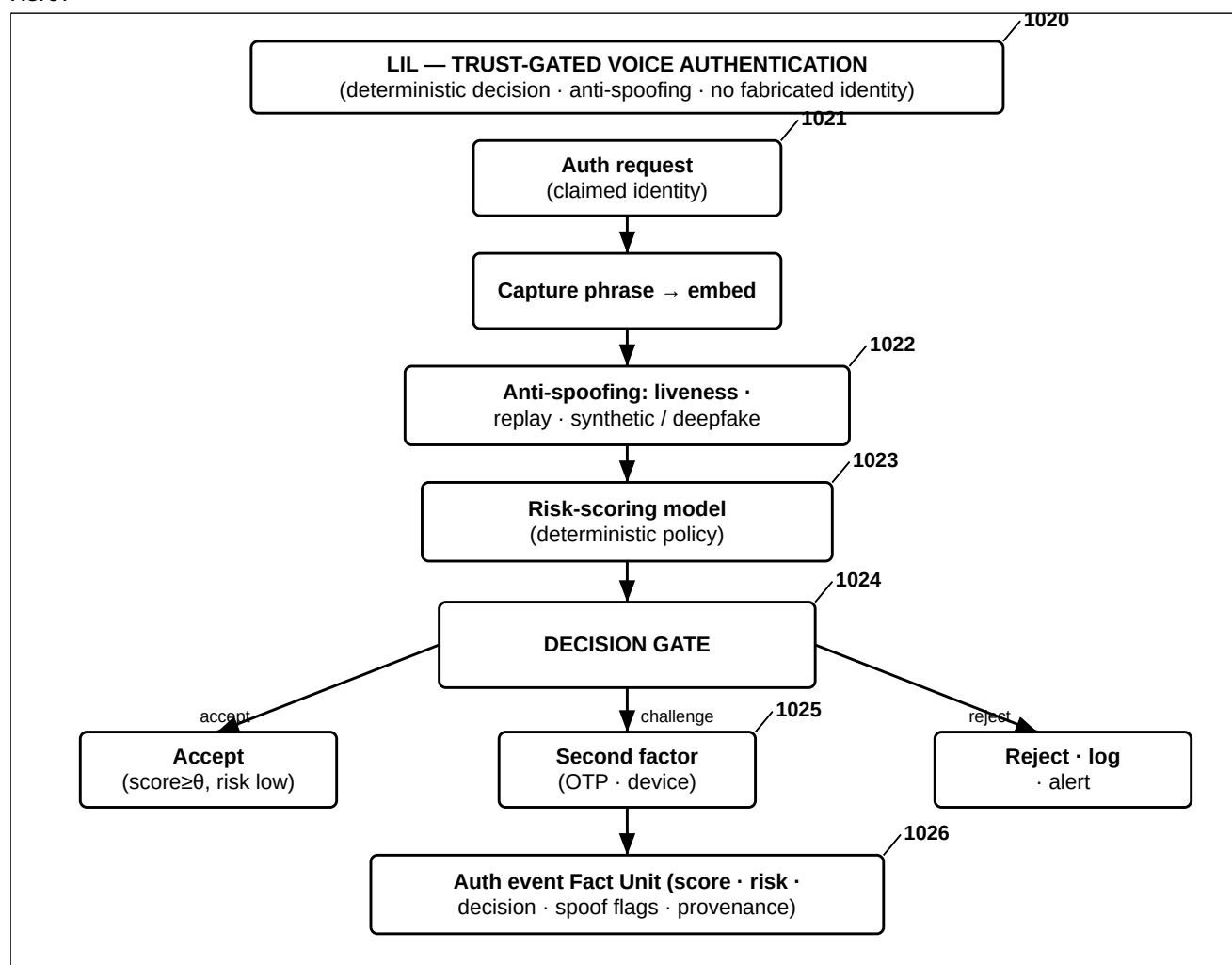


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

FIG. 35

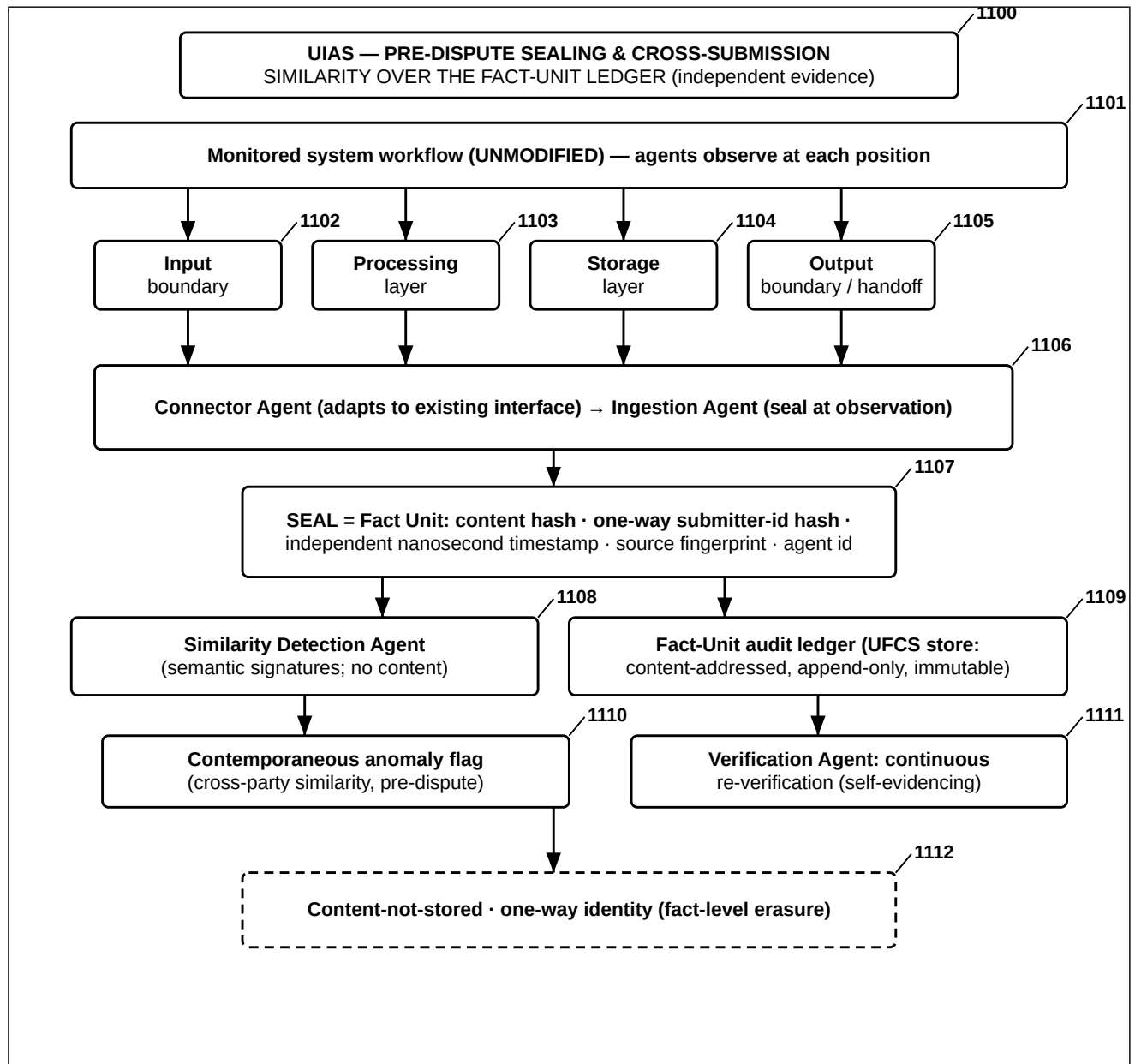


FIG. 35 — UIAS — Pre-Dispute Sealing and Cross-Submission Similarity over the Fact-Unit Ledger

FIG. 36

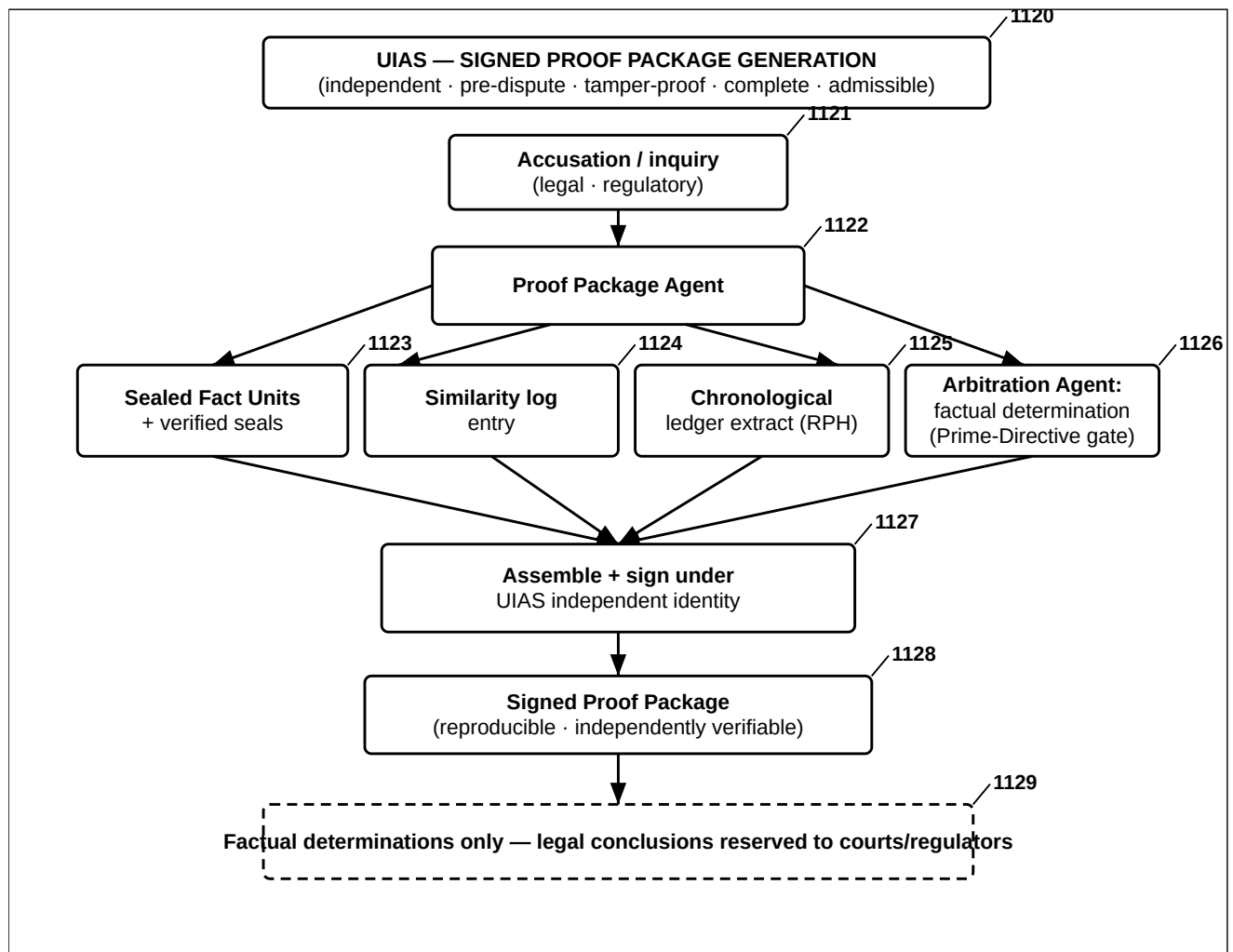


FIG. 36 — UIAS — Signed Proof Package Generation

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Companion drawings volume to the consolidated QueryBook Provisional Patent Application; figures and numerals are drawn from, and identical to, that application. Finalize to 37 CFR 1.84 before filing.